

Forecasting Project Canvas (Individual)

Course: AI for Business Prognosis (AI4BP)

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Forecasting Project Canvas

Individual worksheet. Sketch ONE candidate forecasting project. Print and write on it.
Luftverschmutzung Beijing —

1. Business decision this forecast supports

What decision changes if the forecast is good? Who acts on it?

Soll ein Bauunternehmen/Industriebetrieb in den nächsten 1-3 Tagen den behördlichen Notfallplan aktivieren (Bauarbeiten, Produktion, Lieferverkehr aussetzen/umplanen), weil ein Rotalarm (Peking: prognostizierter Tages-AQI > 200 über mehr als 2 Tage) erwartet wird?

Handelnde: Bauleiter/Site-Manager, Produktionsplaner, Logistik-/Flottendisponenten.

Einzelhandel

Bei guter Prognose können Personal, Material und Lieferungen 1-3 Tage im Voraus umgeplant werden statt kurzfristig zu reagieren.

2. Forecast target

Variable, unit, level / granularity (e.g., weekly units sold per store).

Variable: PM2.5-Konzentration (Proxy für Tages-AQI) Unit: $\mu\text{g}/\text{m}^3$

Level / granularity: stündlich erfasst; 12 Messstationen in Peking (stationsweise, optional zu stadtweitem Mittel aggregiert)

mehrere Stationen und auch weitere Belastungen wie NO₂, SO₂

3. Horizon and frequency

How far ahead? How often is it refreshed? Single- or multi-horizon?

Horizon: 1-72 Stunden (Fokus 48-72h) Frequency: täglich (optional alle 6h)

[] single-horizon [x] multi-horizon

4. Challenge focus (pick at least one)

- [] Long / multi-horizon forecasting
- ☒ Uncertainty quantification
- ☒ Multivariate / multi-series (multi-value) forecasting
- ☒ Exogenous drivers

☐ Intermittent / sparse demand

5. DATA GATE (most important - do this first)

If you cannot get the data, the project cannot proceed. Confirm access early.

Source and link: UCI Machine Learning Repository - Beijing Multi-Site Air-Quality Data, <https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data> (Kaggle-Spiegel: beijing-multisite-airquality-data-set)

Access confirmed? [x] yes [] no

History length: 4 Jahre (03/2013-02/2017) Frequency: stündlich

Exogenous features available? [x] yes [] no

What: Temperatur, Taupunkt, Luftdruck, Niederschlag, Windrichtung, Windgeschwindigkeit

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6. Reference paper (blueprint, NOT full reproduction)

Citation: Zhang, S. et al. (2022). "Prediction of Multi-Site PM2.5 Concentrations in Beijing Using CNN-BiLSTM with CBAM." *Atmosphere*, 13(10), 1719. Interpretable PM2.5 Forecasting for Urban Air Quality: A Comparative Study of Operational Time-Series Models, <https://arxiv.org/html/2603.25495v1>

(Datensatz-Ursprung: Zhang, S. et al. (2017). "Cautionary Tales on Air-Quality Improvement in Beijing." *Proc. Royal Society A*, 473(2205).)

What you take from it:

- ☒ Method / architecture
- ☒ Benchmark number
- ☒ Dataset
- ☐ Metric

Note: use the paper as a blueprint. You are NOT expected to reproduce it fully.

7. Baseline

Which simple baseline and why? (e.g., naive, seasonal naive, mean.)

Baseline: Seasonal naive

Why: PM2.5 hat starken Tagesgang (Verkehr, Heizperiode) und Wochenmuster; seasonal naive ist der Standard-Referenzwert und praktisch kostenlos zu berechnen.

8. Candidate methods

Classical: Prophet (saisonal + exogene Faktoren)

Deep learning: Chornos + PatchTST

Mark the method that is NEW to your team (not your own tutorial method):

[] classical is new [x] deep-learning is new (PatchTST)

Reminder: at least ONE method must differ from your team's tutorial method.

9. Evaluation plan

Metric(s) that fit the challenge: RMSE/MAE; MASE gegen Prophet-Baseline

Hint list – pick what fits:

- Point error: MASE, RMSE
- Uncertainty: automatisch bei Prophet und Chronos

Backtesting scheme:

☐ walk-forward ☐ expanding window ☒ sliding window

Evaluation horizon: 72h,

10. Success criteria and open risks

What does “good” look like? What must be true to call this a success?

Modell schlägt seasonal-naive-Baseline deutlich ($\geq 15\text{-}20\%$ RMSE-Reduktion) auf 24-72h-Horizont.

gutte Performance auch bei Sliding über den Zeitraum

Risiko: sehr komplexes Problem